

Ethics Report *For Smart Reverse*

Vending Machine

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Software Ethics Principles Adopted

Based on the **Software Engineering Code of Ethics and Professional Practice**, our team has adopted the following core ethical principles for the Smart Reverse Vending Machine (RVM) project:

1. PUBLIC Interest (Principle 1)

1.01 - Accept full responsibility for own work

Evidence:

- **FR-25: Comprehensive analytics generation with detailed performance reports**
- **NFR-5.3: Audit logging maintains 12-month records for accountability**

- **Team members explicitly listed with IDs taking ownership**

1.02 - Balance competing interests

Evidence:

- **FR-22: Guest Mode balances inclusivity (public access) with security (rate limiting 5/hour)**
- **FR-8: Bonus system rewards behavior over economic capacity, protecting economically disadvantaged students**

1.03 - Approve software only if it is safe

Evidence:

- **FR-4.2: Minimum 90% accuracy requirement in AI material classification**
- **FR-5: Liquid detection system prevents contamination and machine damage**
- **NFR-2.1: 99% uptime requirement during operational hours**
- **90+ comprehensive test cases documented (TC_Auth_01 through TC_Pts_30)**
- **FR-6.4: Sensor malfunction detection with automatic administrator notifications**
- **NFR-2.3: Hardware failure detection within 5 seconds with error messages**

1.04 - Disclose actual/potential dangers

Evidence:

- **FR-28.2.1, 28.2.2: Capacity alerts at 80%, 90%, 100% thresholds sent to administrators**
- **FR-6.4: Automatic administrator notifications for sensor malfunctions**
- **FR-28.3: Connectivity issues with offline machines notified to admins**
- **FR-20.4: Synchronization status notifications to administrators**

1.05 - Cooperate in efforts to address public concerns

Evidence:

- **FR-25.4: Environmental impact reports showing total weight recycled, CO2 savings**
- **FR-22.5: Guest transactions display environmental impact contribution messages**

1.06 - Be fair and avoid deception in all statements

Evidence:

- **FR-11: Real-time transparent feedback displaying weight, points earned within 3 seconds**
- **FR-7: Clear point calculation formula (10 points/kg) publicly displayed**
- **FR-10.4: Users can export transaction history in CSV format for verification**
- **Competitive Analysis: Honest assessment acknowledging competitors' strengths without exaggeration**

1.07 - Consider factors that diminish access to software

Evidence:

- **FR-22: Guest Mode removes registration barriers for all campus members**
- **NFR-3.2: Multi-language support (English/Arabic) addresses linguistic diversity**
- **NFR-3.3: Readable displays (font ≥ 18 pt, readable from 1.5m distance)**
- **FR-8: Bonus system rewards consistent behavior, not bulk purchases (prevents economic advantage)**
- **FR-7.3: Minimum 50g threshold prevents trivial gaming while remaining accessible**

1.08 - Volunteer professional skills to good causes

Evidence:

- **Entire project addresses campus plastic waste management as environmental cause .**

- **FR-4.4, FR-5.4: Rejection feedback educates users on proper recycling practices .**
- **FR-22.5: Guest deposits display thank you messages showing environmental impact contribution.**

2. CLIENT AND EMPLOYER Interest (Principle 2)

2.01 - Provide service in area of competence

Evidence:

- **Technologies used are standard: React, Raspberry Pi, PostgreSQL, sensors**
- **NFR-4.2: Realistic database capacity (100,000 transactions)**
- **FR-4.2: Achievable 90% accuracy target (not unrealistic 100%)**

2.02 - Don't use software obtained illegally

Evidence:

- **Section 1.2 Ethics Report: "Don't use software obtained illegally - all libraries and frameworks are properly licensed"**
- **Section 3.5 License Management Table: All components properly licensed (MIT, ISC, PostgreSQL License, GPL/LGPL)**
- **Section 2.7 Ethics Report: "All code developed in-house or properly attributed"**

2.03 - Use property only in authorized ways

Evidence:

- **FR-1.2: University email domain validation ensures proper authorization**
- **FR-16: Cafeteria integration via secure API (implies authorization)**

2.05 - Keep information confidential when required

Evidence:

- **NFR-5.1: AES-256 encryption for all data transmission and storage**
- **Section 1.2 Ethics Report: "Respect confidentiality of user data and transaction information"**
- **FR-1.3: Secure authentication with encrypted credentials**

2.06 - Promptly report significant project problems

Evidence:

- **FR-28: Real-time system monitoring dashboard with alerts**
- **FR-28.2.2: Machine malfunction alerts sent to administrators**
- **FR-20.4: Notify administrators of synchronization status and failures**

2.08 - Accept no outside work detrimental to the work they perform for their primary employer

Academic Context

Evidence:

Academic project for CSE251 course - team members are students, no employment conflict, "All team members confirm no conflicting commercial interests in competing recycling solutions"

3 .PRODUCT Quality (Principle 3)

3.01 - Strive for high quality, acceptable cost, and reasonable schedule

Evidence:

- **29 hierarchical functional requirements with clear specifications**
- **Ethics Report Section 2: Systematic documentation of tradeoffs (e.g., comprehensive data vs. privacy, AI accuracy vs. user experience)**

3.02 - Ensure proper and achievable goals

Evidence:

- **NFR-2.1: 99% uptime (achievable, not 100%)**

- **FR-4.2: 90% AI accuracy (realistic, not 100%)**
- **FR-3.4: 2-second QR validation (measurable and achievable)**
- **FR-11.1: 3-second feedback display (specific and testable)**

3.03 - Identify, define, and address real or potential environmental issues

Evidence:

- **Ethical: Comprehensive Ethics Report with 7 major ethical issues analyzed**
- **Economic: FR-8 addresses economic fairness in bonus system**
- **Cultural: NFR-3.2 Multi-language support (English/Arabic)**
- **Legal: Section 3.7 Egyptian Data Protection Law compliance, licensing compliance**
- **Environmental: FR-25.4 Environmental impact reporting, entire project addresses waste management**

3.04 - Ensure that they are qualified for any project on which they work or propose to work by an appropriate combination of education, training, and experience

Evidence:

- **Academic project for CSE251 - Software Engineering course**
- **Section 1.2: "Act within areas of competence by using established technologies"**
- **Technologies used are standard in curriculum: Raspberry Pi, React, databases, sensors**
- **Team of 5 members suggests skill distribution**

3.05 - Ensure an appropriate method is used for any project on which they work or propose to work

Evidence:

- **Section 1.3: "Use appropriate project methodologies following SOLID principles for maintainability"**

- **Section 5: Comprehensive SOLID-compliant subsystem design documented**
- **Hierarchical requirements methodology (29 FRs with sub-requirements)**
- **Systematic testing approach with 90+ test cases**

3.06 - Work to follow professional standards, when available, that are most appropriate for the task at hand

Evidence:

- **ISO 14001 environmental management principles applied (Section 3.10)**
- **AES-256 encryption standard (NFR-5.1, industry standard)**
- **HMAC-SHA256 for QR authentication (NFR, cryptographic standard)**
- **ACM/IEEE-CS Software Engineering Code of Ethics followed throughout**
- **PostgreSQL database standards, React development best practices**

3.07 - Strive to fully understand the specifications for software on which they work

Evidence:

- **29 detailed functional requirements with 200+ sub-specifications**
- **Use Case Descriptions with detailed actor intentions and system responsibilities**
- **Section 8: Detailed interaction scenarios for each use case**
- **Traceability Matrix linking requirements to subsystems**

3.08 - Ensure that specifications for software on which they work have been well documented, satisfy the users' requirements and have appropriate approvals

Evidence:

- **Well documented: Comprehensive documentation (FRs, NFRs, Use Cases, Test Cases)**
- **Satisfy user requirements: Requirements address students, guests, admins, cafeteria**

- **Appropriate approvals:** documented formal approval from university, cafeteria partners, or student representatives

3.09 - Ensure realistic quantitative estimates of cost, scheduling, personnel, quality and outcomes

Evidence:

- **NFR-1: Performance metrics (2-second validation, 3-second receipt, 50 items/hour)**
- **NFR-2.1: 99% uptime during operational hours**
- **NFR-4.2: Database capacity for 100,000 transactions**
- **FR-6.2: Weight accuracy $\pm 5g$ tolerance**
- **FR-4.2: 90% material classification accuracy**

3.10 - Ensure adequate testing, debugging, and review

Evidence:

- **90+ comprehensive test cases documented (Sections 11, Use Cases 1-6)**
- **Test coverage: Authentication (30 cases), Transaction (30 cases), QR Code (30 cases), Notifications (30 cases), Points (30 cases)**
- **Validation testing requirements specified for each functional requirement**

3.11 - Ensure adequate documentation including problems and solutions

Evidence:

- **Detailed functional requirements (29 FRs)**
- **Non-functional requirements, use cases, test cases, ethics report provided**
- **Traceability Matrix, SOLID subsystem design documentation**
- **Use Case interaction scenarios with detailed system responses**

3.12 - Work to develop software that respects privacy

Evidence:

- **NFR-5.1: AES-256 encryption for data transmission and storage**
- **FR-22: Guest Mode option for zero-data participation**
- **Section 2.1: Data retention policies (minimum 2 months, not indefinite storage)**
- **FR-1: User consent through registration with clear privacy policies**
- **Section 1.3: "Develop software that respects privacy through encrypted data transmission and storage"**

3.13 - Be careful to use only accurate data derived by ethical and lawful means

Evidence:

- **FR-6.1, 6.2: Calibrated weight sensors with $\pm 5g$ accuracy tolerance**
- **FR-6.3: Automatic weekly calibration for measurement accuracy**
- **FR-4.4, FR-5.4: Accurate logging of classification and rejection results**
- **No deceptive data collection practices**
- **FR-1.2: Email domain validation ensures legitimate user data**

3.14 - Maintain the integrity of data, being sensitive to outdated or flawed occurrences

Evidence:

- **FR-20: Automatic data synchronization when connectivity restored**
- **FR-21: Conflict resolution system for duplicate transactions and timestamp conflicts**
- **FR-6.4: Sensor malfunction detection and reporting**
- **FR-19.2: Offline mode stores transactions locally with integrity checks**
- **NFR-2.2: Automatic data backup every 24 hours**

3.15 - Treat all forms of software maintenance with the same professionalism as new development

Evidence:

- **FR-27.8: Machine maintenance history and schedules tracked**
- **FR-6.3: Automatic weekly sensor calibration**
- **FR-28: Comprehensive system monitoring and alerting**
- **Section 5 Subsystem Design: SensorCalibrator class with dedicated calibration responsibility**
- **Admin dashboard includes maintenance scheduling features**

4 . JUDGMENT Integrity (Principle 4)

4.01 - Temper all technical judgments by need to support human values

Evidence:

- **FR-22: Guest Mode prioritizes human inclusivity over system simplicity**
- **FR-8: Bonus system rewards behavior patterns, not economic capacity (protects disadvantaged students)**
- **Section 2.4: "Balanced gamification rewarding behavior over volume"**
- **Environmental impact prioritized over profit/efficiency in validation system design**

4.02 - Only endorse documents prepared under your direction or within competence

Evidence:

- **All team members listed with IDs: Yasseen (23101608), Mohamed M. (23101594), Ahmed (23101545), Marawan (23101599), Mohamed A. (23101899)**
- **Technical specifications based on documented competencies: Raspberry Pi, React, sensors (within CSE251 curriculum)**
- **Section 1.2: "Act within areas of competence"**

4.03 - Maintain professional objectivity in evaluations

Evidence:

- FR-25.6: Objective rejection analytics for quality monitoring
- FR-4.4: Unbiased classification logging (all results recorded, not selective)
- Competitive Analysis (Section 9): Fair assessment of competitors (Recyclobekia, Dawarha, ZeroPrime, Bekia) citing their strengths

4.04 - Not engage in deceptive financial practices such as bribery, double billing, or other improper financial practices

Evidence:

- FR-7: Transparent point system (10 points/kg publicly displayed)
- FR-17.2: Atomic transaction processing prevents double-spending
- Clear redemption rates (1 point = 1 EGP per TR-ISS-02)
- No hidden fees or costs

4.05 - Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped

Evidence:

- Competitive Analysis (Section 9): Openly cites and discusses competitors without hiding their existence
- Section 3.3: All competitor information sourced from publicly available materials, properly cited
- Technology attributions provided (Raspberry Pi, React, libraries)
- Section 1.4: "Disclose conflicts of interest in competitive analysis and technology selection"
- Academic context transparent (CSE251 course project)

4.06 - Refuse to participate in or support work that violates applicable laws or this Code

Evidence:

- Section 3.2-3.5: Proper licensing compliance (MIT, Apache 2.0, PostgreSQL)
- Section 3.7: Proactive Egyptian Data Protection Law compliance measures
- NFR-5: Security standards following industry best practices
- No malicious code, no unlicensed software, no privacy violations
- Team adopted ACM/IEEE-CS Code as framework (refuses to violate it)

5. MANAGEMENT

5.01 - Ensure good management for any project on which you work

Evidence:

- FR-23: Multi-machine management system across campus
- FR-27: Comprehensive administrator controls for machine management
- FR-28: Real-time system monitoring and alerting
- NFR-2.1: 99% uptime SLA with clear quality target
- Section 2.5: Risk mitigation through offline capability and redundancy

5.02 - Ensure that software engineers are informed of standards before being held to them

Evidence:

- NFR-5: Security standards clearly documented (AES-256, HMAC-SHA256)
- FR-4.2: 90% accuracy standard specified upfront
- All performance requirements specified in NFR section before implementation

- Section 1: Ethics principles explicitly adopted and communicated to team

5.03 - Ensure that software engineers know the employer's policies and procedures for protecting passwords, files and information that is confidential to the employer or confidential to others

Evidence:

- NFR-5.1: Encryption policies documented (AES-256)
- NFR-5.2: Session management procedures (5-minute timeout)
- NFR-5.3: Audit logging procedures (12-month retention)
- FR-1.3: Secure authentication protocols with encrypted credentials
- Section 2.1: Clear privacy policies and data protection procedures

5.04 - Assign work only after taking into account appropriate contributions of education and experience balanced with desires to further that education and experience

Evidence:

- Role separation: Registered users, guests, administrators (appropriate to skill levels)
- FR-29: User Management shows admin-only privileges for sensitive operations
- Team structure with 5 members suggests appropriate work distribution
- Academic context: Project within CSE251 curriculum scope

5.05 - Ensure realistic quantitative estimates of cost, scheduling, personnel, quality and outcomes on any project on which they work and provide an uncertainty assessment of these estimates

Evidence:

- NFR-1: Specific performance metrics (2 seconds, 3 seconds, 50 items/hour)
- NFR-2.1: 99% uptime (acknowledges 1% acceptable downtime)

- FR-4.2: 90% accuracy (acknowledges 10% error margin)
- FR-6.2: $\pm 5g$ tolerance (uncertainty explicitly stated)

5.09 - Ensure that there is a fair agreement concerning ownership of any software, processes, research, writing, or other intellectual property to which a software engineer has contributed

Evidence:

- All code developed in-house with implied team/university ownership
- Proper attribution of third-party libraries (Section 3.5)
- Academic project context implies standard university IP policies apply
- Section 2.7: "Original work: All designs, UI mockups, system architecture developed in-house"

5.10 - Provide for due process in hearing charges of violation of an employer's policy or of this Code

Evidence:

- FR-29.6: Administrator handles user disputes and support tickets (appeals mechanism)
- FR-21: Systematic conflict resolution for transaction disputes
- Section 2.3: "Appeal mechanism through administrator support"
- User can challenge rejected items through admin support

5.11 - Not ask a software engineer to do anything inconsistent with this Code

Evidence:

- Team collectively adopted ACM/IEEE-CS ethics code as framework
- Ethics Report demonstrates no requests for unethical practices
- Systematic ethical evaluation throughout design (Section 2)

- No evidence of pressure to compromise privacy, security, or environmental goals

5.12 - Not punish anyone for expressing ethical concerns about a project

Evidence:

- Comprehensive Ethics Report shows open discussion of ethical challenges
- Section 4: "Ethical Challenges and Resolutions" documents concerns raised
- Collaborative team structure (5 members) implies safe environment
- Academic supervision provides protection for raising concerns

6. PROFESSION

6.01 - Help develop an organizational environment favorable to acting ethically

Evidence:

- Project promotes campus-wide environmental responsibility
- FR-22: Inclusive guest access fosters positive participation culture
- Transparent point system builds trust in gamification
- Section 2.2: "Inclusive system serving entire campus community"

6.02 - Promote public knowledge of software engineering

Evidence:

- FR-4.4, FR-5.4: User education through rejection feedback on proper recycling
- Project demonstrates practical application of IoT, AI, mobile development
- Promotes understanding of sustainable technology solutions
- Section 1.8: "Participate in lifelong learning about environmental impact and recycling technologies"

6.03 - Extend software engineering knowledge by appropriate participation in professional organizations, meetings and publications

Evidence:

- Academic project for CSE251 course (professional academic environment)
- Comprehensive Ethics Report serves as documentation/publication of practices
- Project report contributes to body of knowledge on sustainable software engineering
- Academic course participation IS professional development

6.04 - Support, as members of a profession, other software engineers striving to follow this Code**Evidence:**

- Section 1.7: "Encourage colleagues to adhere to this Code throughout development"
- Team of 5 members working collaboratively
- Ethics code adopted as team standard
- Mutual support in addressing ethical challenges evident in Ethics Report

6.05 - Not promote their own interest at the expense of the profession, client or employer**Evidence:**

- Fair competitive analysis citing competitors positively (Section 9)
- Project serves university/environmental interests, not personal profit
- Transparent methodology without deceptive practices
- Section 1.7: "Give credit where deserved in competitive analysis"

6.06 - Obey all laws governing their work, unless, in exceptional circumstances, such compliance is inconsistent with the public interest

Evidence:

- Section 3.5: Proper software licensing compliance
- Section 3.7: Proactive Egyptian Data Protection Law alignment
- University authorization protocols for campus deployment
- No evidence of law violations

6.07 - Be accurate in stating the characteristics of software on which they work, avoiding not only false claims but also claims that might reasonably be supposed to be speculative, vacuous, deceptive, misleading, or doubtful

Evidence:

- Honest accuracy claims: 90% not 100% (FR-4.2)
- Realistic uptime: 99% not 100% (NFR-2.1)
- Limitations acknowledged ($\pm 5g$ tolerance, 10% error margin)
- No exaggerated environmental impact claims
- Section 1.4: "Remain objective when evaluating system performance"

6.08 - Take responsibility for detecting, correcting, and reporting errors in software and associated documents on which they work

Evidence:

- FR-28: Comprehensive system monitoring and alerts
- FR-6.4: Sensor malfunction detection and reporting
- FR-21: Conflict detection and resolution mechanisms
- FR-25.6: Quality monitoring analytics for continuous improvement
- Section 2.5: "Comprehensive monitoring to protect users and data integrity"

6.09 - Ensure that clients, employers, and supervisors know of the software engineer's commitment to this Code of ethics, and the subsequent ramifications of that commitment

Evidence:

- Section 1: "Software Ethics Principles Adopted" explicitly states adoption of ACM/IEEE-CS Code
- Comprehensive Ethics Report submitted to course supervisor
- University receives ethics documentation as project deliverable
- Ethics Report IS the communication of commitment to stakeholders

6.10 - Avoid associations with businesses and organizations which are in conflict with this Code

Evidence: Reputable technology partners chosen (Raspberry Pi, React - all open source/ethical) , "Cafeteria partner [Name] evaluated for ethical business practices including fair labor, transparent pricing, and food safety standards"

6.11 - Recognize that violations of this Code are inconsistent with being a professional software engineer

Evidence:

- Systematic ethical evaluation throughout Ethics Report
- Section 4: "Ethical Challenges and Resolutions" shows awareness of potential violations
- Proactive identification and mitigation of ethical risks
- Team commitment to Code demonstrated through comprehensive analysis

7. COLLEAGUES Support (Principle 7)

7.01 - Encourage colleagues to adhere to this Code

Evidence:

- Section 1.7: "Encourage colleagues to adhere to this Code throughout development"
- Team collectively adopted ACM/IEEE-CS Software Engineering Code of Ethics
- Ethics framework applied to all design decisions by entire team
- Mutual accountability established

7.02 - Assist colleagues in professional development

Evidence:

- 5-member collaborative team: Yasseen, Mohamed M., Ahmed, Marawan, Mohamed A.
- Knowledge sharing in IoT, AI validation, mobile development technologies
- Section 1.7: "Assist in professional development through collaborative team structure"
- Academic project context provides learning opportunities

7.03 - Credit fully the work of others and refrain from taking undue credit

Evidence:

- Section 3.3: All competitor information properly cited (Recyclobekia, Dawarha, ZeroPrime, Bekia)
- Section 3.2-3.5: All technology sources attributed (Raspberry Pi, React, libraries)
- Section 1.7: "Give credit where deserved in competitive analysis and technology references"
- No plagiarism of designs or implementations

7.05 - Give a fair hearing to the opinions, concerns, or complaints of a colleague

Evidence:

- **Comprehensive Ethics Report shows consideration of multiple perspectives**
- **Section 2: Systematic evaluation of alternatives demonstrates open discussion (e.g., Guest Mode alternatives, validation alternatives)**
- **Collaborative approach to ethical challenges evident**
- **Multiple viewpoints considered in design decisions**

7.06 - Assist colleagues in being fully aware of current standard work practices including policies and procedures for protecting passwords, files and other confidential information, and security measures in general

Evidence:

- **NFR-5: Security standards documented and shared (AES-256, HMAC-SHA256)**
- **Team collaboration implies shared understanding of encryption practices**
- **Authentication and data protection procedures clearly specified**
- **Section 2.1: Privacy policies and security measures documented for all**

7.07 - Not unfairly intervene in the career of any colleague; however, concern for the employer, the client or public interest may compel software engineers, in good faith, to question the competence of a colleague

Evidence:

- **FR-8: Fair gamification system rewards behavior, not individual advantage**
- **FR-13: College competition aggregates contributions equitably**
- **No mechanisms that unfairly disadvantage any participants**

- Section 2.4: "Do not unfairly interfere with another's career but protect public interest"

8. SELF Development (Principle 8)

8.01 - Further knowledge of developments in software engineering

Evidence:

- **Project involves learning IoT sensor integration, AI-powered classification, mobile app development**
- **Real-time systems, cloud synchronization, offline mode development**
- **Section 1.8: "Stay current with developments in IoT, AI validation, and sustainable technologies"**
- **Advanced technologies: QR authentication, material classification, liquid detection**

8.02 - Improve ability to produce accurate, quality software

Evidence:

- **Section 1.8: "Improve ability to create high-quality software through SOLID design patterns"**
- **90+ test cases for reliability (comprehensive testing approach)**
- **Clear NFR targets for quality: 99% uptime, 90% accuracy, specific response times**
- **Comprehensive validation systems (FR-4, FR-5) ensure safety**

8.03 - Improve their ability to produce accurate, informative, and well-written documentation

Evidence:

- **Comprehensive project documentation: 29 FRs with 200+ sub-requirements**
- **Detailed NFRs, Use Cases with interaction scenarios, Test Cases**
- **Ethics Report with systematic analysis**
- **Traceability Matrix, SOLID subsystem design documentation**
- **Section 8: Detailed interaction scenarios demonstrate documentation skill**

8.04 - Improve their understanding of the software and related documents on which they work and of the environment in which they will be used

Evidence:

- **Campus context analysis (understanding university environment)**
- **User needs assessment: students, guests, faculty, administrators**
- **Cafeteria integration requirements (understanding ecosystem)**
- **Environmental impact considerations throughout design**
- **Competitive Analysis (Section 9) shows understanding of market environment**

8.05 - Improve their knowledge of relevant standards and the law governing the software and related documents on which they work

Evidence:

- **ISO 14001 environmental principles applied**
- **Encryption standards (AES-256, HMAC-SHA256)**
- **Proactive Egyptian Data Protection Law alignment (Section 3.7)**

8.06 - Improve their knowledge of this Code, its interpretation, and its application to their work

Evidence:

- **Systematic application of ACM/IEEE-CS Code throughout project**
- **Comprehensive Ethics Report demonstrates deep understanding**
- **Code principles explicitly mapped to design decisions (Section 2)**
- **Section 1.8: "Participate in lifelong learning" includes learning ethics**

8.07 - Not give unfair treatment to anyone because of any irrelevant prejudices

Evidence:

- **FR-22: Inclusive guest mode (no discrimination by registration status)**
- **NFR-3.2: Multi-language support shows cultural awareness**
- **Fair gamification preventing economic discrimination**

8.08 - Not influence others to undertake any action that involves a breach of this Code

Evidence:

- **Team adherence to ethics code throughout**
- **No encouragement of gaming system or abuse**
- **FR-22: Rate limiting and abuse prevention measures designed ethically**
- **Proper licensing compliance (no pressure to use pirated software)**
- **Section 2.7: "No proprietary code copying"**

8.09 - Recognize when violations of this Code are not being addressed

Evidence:

- Systematic ethical evaluation shows awareness of Code importance
- Section 4: Proactive identification of ethical challenges
- Commitment to ethics demonstrated through comprehensive analysis

COMPREHENSIVE GAP SUMMARY

#	Principle	Code	Description	Status	Priority	Reason/Recommendation
1	CLIENT/EMPLOYER	2.04	Ensure documents are approved by authorized persons	NOT ADDRESSED	HIGH	Add stakeholder approval log with dates
2	CLIENT/EMPLOYER	2.07	Report significant social concerns to employer/client	NOT ADDRESSED	HIGH	Create social concern reporting procedure
3	CLIENT/EMPLOYER	2.08	Accept no outside work detrimental to primary employer	PARTIAL/N/A	LOW	Academic context - no employment. Add brief conflict disclosure statement
4	CLIENT/EMPLOYER	2.09	Promote no interest adverse to employer unless higher	NOT ADDRESSED	MEDIUM	Document conflict resolution framework

ethical concern						
5	PRODUCT	3.08	Ensure specifications have appropriate approvals	PARTIAL	MEDIUM	Well documented, satisfies users, but missing formal approval sign-offs
6	PROFESSION	6.10	Avoid associations with businesses in conflict with Code	PARTIAL	MEDIUM	Tech partners ethical, but cafeteria partner ethics not documented
7	PROFESSION	6.12	Express concerns when Code violations detected	NOT ADDRESSED	HIGH	Create internal ethics violation procedure
8	PROFESSION	6.13	Report significant violations to appropriate authorities	NOT ADDRESSED	HIGH	Establish whistleblowing policy
9	COLLEAGUES	7.04	Review work of others objectively and document properly	NOT ADDRESSED	MEDIUM	Document peer review process
10	COLLEAGUES	7.08	Consult experts outside your competence area	NOT ADDRESSED	MEDIUM	Add expert consultation log
11	SELF	8.05	Improve knowledge of relevant	PARTIAL		Good on tech standards, add Egyptian recycling/consumer

			standards and law		MEDIUM	protection laws
12	SELF	8.07	Not give unfair treatment due to irrelevant prejudices	PARTIAL	MEDIUM	Inclusive features present, add explicit anti-discrimination policy
13	SELF	8.09	Recognize violations inconsistent with being professional	PARTIAL	MEDIUM	Awareness shown, add personal accountability commitment

Application of Recognized Ethics Principles

Privacy and Data Protection

Ethical Issue: The system collects personal information (student ID, email, transaction history, location data) and generates unique QR codes linked to user accounts.

Ethical Alternatives & Consequences:

Alternative	Consequences	Our Choice
A. Collect minimal data only	✓ Enhanced privacy ✗ Limited gamification features	✗ Not chosen

	✗ Reduced analytics capability	
B. Collect comprehensive data with encryption	✓ Full feature implementation ✓ Strong security measures ✗ Requires robust infrastructure	☒ CHOSEN
C. Anonymous system (no registration)	✓ Maximum privacy ✗ No personalized rewards ✗ No competition/leaderboards	☒ Not chosen

Action Taken:

- **Implemented AES-256 encryption** for all data transmission and storage (NFR-5.1)
- **Secure authentication** with encrypted credentials and password recovery
- **Guest Mode option** allowing anonymous participation without data collection
- **User consent** through registration process with clear privacy policies
- **Data retention policies** with transaction history stored for minimum 2 months (FR-10.2)
- **Audit logging** maintains 12-month record for security purposes (NFR-5.3)

Ethical Justification:

We balanced personalization benefits (rewards, rankings, engagement) with privacy protection through strong encryption, guest mode, and transparent data practices per **Principle 3.12** (respect privacy of those affected by software).

Inclusivity and Accessibility

Ethical Issue: A registration-only system would exclude faculty, staff, visitors, and students who choose not to register, creating environmental participation barriers.

Ethical Alternatives & Consequences:

Alternative	Consequences	Our Choice
A. Registered students only	✓ Simplified system ✗ Excludes campus community ✗ Lower environmental impact	✗ Not chosen
B. Guest Mode + Registered Mode	✓ Inclusive participation ✓ Higher recycling rates ✗ Additional complexity	☑ CHOSEN
C. Public access with abuse risks	✓ Maximum access ✗ Gaming/abuse vulnerabilities ✗ System integrity concerns	✗ Not chosen

Action Taken:

- **Guest Mode implementation** (FR-22) with prominent "Guest" button on RVM interface
- **Same validation standards** for guest deposits (material classification, liquid detection)
- **Thank you messaging** showing environmental impact contribution
- **Rate limiting** (max 5 deposits/hour) to prevent abuse while maintaining accessibility
- **Separate analytics tracking** for guest vs. registered contributions
- **Multi-language support** (English/Arabic) per NFR-3.2

Ethical Justification:

Following **Principle 1.07** (consider factors that diminish access to software), we created an inclusive system serving the entire campus community while maintaining system integrity through balanced abuse prevention measures.

+ Environmental Impact Validation

Ethical Issue: Without proper validation, the system could accept non-recyclable materials, contaminating recycling streams and undermining environmental goals.

Ethical Alternatives & Consequences:

Alternative	Consequences	Our Choice
A. Trust users completely	✓ Simple implementation ✗ High contamination risk ✗ Reduced recycling quality	✗ Not chosen
B. Manual inspection	✓ High accuracy ✗ Labor intensive ✗ Not scalable	✗ Not chosen
C. AI-powered automated validation	✓ Scalable solution ✓ Real-time feedback ✗ Requires ML development	☑ CHOSEN

Action Taken:

- **MaterialClassifier** achieving minimum 90% accuracy (FR-4.2) using AI-powered sensors
- **LiquidDetector** preventing liquid-contaminated bottles from entering recycling stream (FR-5)
- **Automatic waste diversion** for non-plastic materials to separate compartment (FR-4.3)
- **Clear rejection feedback** educating users on proper recycling practices
- **Rejection analytics** (FR-25.6) identifying patterns for targeted education campaigns
- **Quality monitoring dashboard** for administrators to track validation system performance

Ethical Justification:

Per **Principle 1.03** (approve software only if it is safe) and **Principle 3.03** (face up to environmental issues), we prioritized recycling quality over convenience to ensure genuine environmental impact rather than creating a "feel-good" system that actually harms recycling efforts.

Fairness in Gamification

Ethical Issue: Point systems and leaderboards could disadvantage students with limited financial resources to purchase recyclables or create unhealthy competition.

Ethical Alternatives & Consequences:

Alternative	Consequences	Our Choice
A. No gamification	✓ No competition concerns ✗ Lower engagement ✗ Reduced participation	✗ Not chosen
B. Unlimited deposits	✓ Maximum participation ✗ Gaming through bulk purchases ✗ Unfair advantages	✗ Not chosen
C. Balanced bonus system	✓ Encourages consistent participation ✓ Rewards behavior patterns ✗ Moderate complexity	☑ CHOSEN

Action Taken:

- **Minimum deposit threshold** of 50g to earn points (FR-7.3) preventing trivial gaming
- **Bonus for consistency** (20% first deposit of day, 30% weekly streak) rewarding behavior over volume
- **College-level competition** aggregating contributions across economic backgrounds
- **Points expiration** after 15 days inactivity (FR-24) preventing hoarding
- **Guest contributions** count toward campus totals without individual rewards
- **Transparent point calculation** (10 points/kg base rate) displayed in real-time

Ethical Justification:

Following **Principle 1.02** (balance competing interests) and **Principle 7.07** (do not unfairly interfere with another's career but protect public interest), we designed gamification to motivate genuine environmental behavior rather than financial competition.

 System Reliability and Safety

Ethical Issue: Machine malfunctions could cause user injury, data loss, or environmental contamination if waste compartments are not managed properly.

Ethical Alternatives & Consequences:

Alternative	Consequences	Our Choice
A. Basic error handling	✓ Simple implementation ✗ Data loss risks ✗ Safety concerns	✗ Not chosen
B. Comprehensive monitoring + offline capability	✓ Data integrity ✓ Continuous operation ✗ Higher complexity	☑ CHOSEN

C. Cloud-only operation

- ✓ Centralized control
- ✗ Complete failure during outages
- ✗ User frustration

✗ Not chosen

Action Taken:

- **99% uptime requirement** during operational hours (NFR-2.1)
- **Offline mode operation** with local transaction storage (FR-19)
- **Automatic data synchronization** when connectivity restored (FR-20)
- **Capacity alerts** at 80%, 90%, 100% thresholds (FR-28.2.1, 28.2.2) preventing overflow
- **Sensor malfunction detection** with automatic administrator notifications (FR-6.4)
- **Automatic weekly calibration** ensuring measurement accuracy (FR-6.3)
- **Graceful degradation** for hardware failures with clear user communication

Ethical Justification:

Per **Principle 1.03** (approve software only if it is safe) and **Principle 2.06** (promptly report problems with project), we implemented comprehensive monitoring and redundancy to protect users, data integrity, and environmental outcomes.

+ Transparency and Accountability

Ethical Issue: Users need to understand how their data is used, how points are calculated, and what happens to collected materials.

Ethical Alternatives & Consequences:

Alternative	Consequences	Our Choice
A. Black-box system	✓ Simpler documentation ✗ User distrust	✗ Not chosen

✗ Accountability concerns		
B. Full transparency with educational content	✓ User trust ✓ Educational value ✗ Requires comprehensive documentation	 CHOSEN
C. Partial disclosure	✓ Moderate effort ✗ Incomplete user understanding ✗ Potential distrust	 Not chosen

Action Taken:

- Real-time feedback displaying weight, points earned, and bonuses within 3 seconds (FR-11)
- Detailed transaction history with export capability (FR-10)
- Transparent point calculation formula (10 points/kg × material factor) clearly documented
- Environmental impact reporting showing CO2 savings and weight recycled (FR-25.4)
- Rejection reason logging with clear user instructions (FR-4.4, FR-5.4)
- Guest impact display showing contribution to campus sustainability (FR-22.5)
- Comprehensive audit trails for all transactions (NFR-5.3)

Ethical Justification: Following Principle 1.06 (be fair and avoid deception in all statements) and Principle 3.11 (ensure adequate documentation), we prioritized transparency to build trust and educate users about their environmental impact.

Intellectual Property and Licensing

Ethical Issue: The project uses various third-party libraries, frameworks, and potentially copyrighted technologies that must be properly licensed.

Ethical Alternatives & Consequences:

Alternative	Consequences	Our Choice
A. Use unlicensed software	✓ No licensing costs ✗ Legal violations ✗ Ethical breaches	✗ Not chosen
B. Open-source only	✓ Clear licensing ✗ Limited technology options ✗ May lack required features	✗ Not chosen
C. Properly licensed mix	✓ Best technology fit ✓ Legal compliance ✗ Requires careful management	✓ CHOSEN

Action Taken:

- Open-source hardware: Raspberry Pi (official licensing)
- Properly licensed libraries: All React components, Tailwind CSS, Lucide icons, Recharts under permissive licenses (MIT, Apache 2.0)
- Cloud storage: Using documented window.storage API with proper attribution
- No proprietary code copying: All code developed in-house or properly attributed
- Competitive analysis disclosure: Properly cited all competitor information (Recyclobekia, Dawarha, ZeroPrime, Bekia)
- Academic integrity: Project report properly references Software Engineering Code of Ethics

Ethical Justification: Per Principle 2.02 (don't use software obtained illegally) and Principle 2.03 (only use property in authorized ways), we ensured all software components are properly licensed, protecting both our project and respecting intellectual property rights of others

Documentation Sources, References, and Legal Compliance

Primary Ethics Framework

ACM/IEEE-CS Software Engineering Code of Ethics and Professional Practice
(Version 5.2)

Used as primary ethical framework for all design decisions

Referenced throughout requirements analysis and system design

Applied to evaluate trade-offs between competing interests

❖ Technical Documentation Sources

- Raspberry Pi Official Documentation (<https://www.raspberrypi.org/documentation/>)
- Hardware specifications and GPIO interfacing
- License: Creative Commons Attribution-ShareAlike 4.0
- React Documentation (<https://react.dev/>)
- Component architecture and hooks implementation
- License: MIT License
- Tailwind CSS Documentation (<https://tailwindcss.com/docs>)
- Styling framework for mobile application
- License: MIT License
- Three.js Documentation (<https://threejs.org/docs/>) (r128)
- Potential 3D visualization components
- License: MIT License

❖ Competitive Analysis Sources

Recyclobekia: Public website and media coverage (properly cited)

Dawarha: Company website, Nestlé partnership announcements (publicly available)

ZeroPrime Technologies (CanBank): Public documentation and deployment information

Bekia: Platform documentation and service descriptions

Traditional Campus Programs: AUC, GUC, Cairo University public sustainability reports

Citation Compliance: All competitor information sourced from publicly available materials; no proprietary or confidential information used.

❖ Academic References

Software Engineering Course Materials (CSE251 - Lecture slides on Software Ethics)

Used as foundational knowledge for ethics principles

Applied 8 core principles framework to project decisions

❖ License Management

Component	License	Usage	Compliance
Raspberry Pi OS	Debian-based (GPL/LGPL)	Operating system for RVM hardware	✓ Open source, properly attributed
React	MIT License	Mobile app framework	✓ Commercial use permitted

Tailwind CSS	MIT License	UI styling	✓ Attribution maintained
Lucide Icons	ISC License	Icon library	✓ Permissive license
Recharts	MIT License	Data visualization	✓ Commercial use permitted
Python (Sensors)	PSF License	Sensor interfacing	✓ Open source compliant
QR Code Libraries	Varies (MIT/Apache 2.0)	QR generation/scanning	✓ All permissively licensed
Database (PostgreSQL)	PostgreSQL License	Data storage	✓ Open source, liberal license

No GPL Contamination: Careful component selection avoids GPL contamination of proprietary student-facing application while using GPL components appropriately in isolated hardware interface layer.

❖ Copyright Compliance

- No copyrighted material reproduced in project documentation without permission
- Fair use applied for competitive analysis (factual descriptions only)
- Original work: All designs, UI mockups, system architecture developed in-house
- Proper attribution: All referenced technologies credited appropriately

❖ Privacy and Legal Compliance (Egyptian Context)

Data Protection Considerations:

- ❖ Egyptian Data Protection Law Draft (pending legislation) - proactive compliance measures:
 - User consent collected during registration
 - Data minimization principle applied
 - Right to deletion implemented (account deactivation per FR-29.3)
 - Transparent data usage policies

University Compliance:

- Email domain validation ensures legitimate student verification
- University collaboration for cafeteria integration (authorized partnerships)
- Campus deployment permissions required before physical installation

Consumer Protection:

- Transparent point system with clear conversion rates (10 points/kg)
- No deceptive practices in gamification or reward mechanisms
- Accurate environmental impact calculations based on verified recycling data

❖ Accessibility and Universal Design

- Multi-language support (English/Arabic) per NFR-3.2 addressing linguistic diversity (global impact)
- Readable displays (font size ≥ 18 pt, readable from 1.5m) per NFR-3.3

- Guest Mode eliminating registration barriers for inclusive participation
- Clear visual feedback with icons and animations for diverse literacy levels

❖ Security Standards Compliance

- AES-256 encryption (industry standard) for data transmission and storage
- HMAC-SHA256 signatures for QR code authentication preventing tampering
- Session management with automatic logout after 5 minutes inactivity (NFR-5.2)
- Rate limiting preventing abuse while maintaining accessibility

❖ Environmental Standards Alignment

- ISO 14001 principles (Environmental Management) informing system design
- Circular Economy concepts applied through closed-loop reward system
- Waste hierarchy adherence: Prevention > Reuse > Recycle hierarchy respected
- Quality over quantity: Validation system ensures genuine recycling outcomes

❖ . Ethical Challenges and Resolutions

Challenge 1: Balancing Gamification with Environmental Authenticity

Issue: Risk of students purchasing recyclables just for points rather than changing consumption behavior.

Resolution:

- Bonus system rewards consistency (streaks) over volume
- Points expiration prevents hoarding
- Educational messaging emphasizes behavior change
- Rejection analytics identify non-authentic patterns

Challenge 2: Guest Privacy vs. System Security

Issue: Allowing anonymous guest access creates abuse potential while mandatory registration excludes participants.

Resolution:

- Rate limiting (5 deposits/hour/machine) balances access and security
- Same validation standards prevent exploitation
- Separate analytics maintain data integrity
- Conversion pathway (app download QR) encourages registration

Challenge 3: AI Validation Accuracy vs. User Experience

Issue: 90% accuracy means 10% error rate potentially frustrating users.

Resolution:

- Clear feedback explaining rejection reasons
- Appeal mechanism through administrator support (FR-29.6)
- Continuous improvement through rejection analytics

- User education prioritized over convenience

Challenge 4: Data Collection vs. Privacy

Issue: Comprehensive gamification requires personal data collection.

Resolution:

- Transparent privacy policies during registration
- Strong encryption (AES-256) protecting all data
- Guest mode offering zero-data alternative
- Retention limits (2 months minimum, not indefinite)

6. Team Contributions and Collaboration (Updated)

6.1 Ethics Principles Framework (Section 1)

Public, Client & Employer – Ahmed Khaled

- Applied Principle 1 (Public) and Principle 2 (Client & Employer)
- Integrated ethical responsibilities regarding safety, transparency, legal compliance, and stakeholder interest
- Provided evidence mapping for requirements related to public interest and employer-client obligations

Product, Self – Marawan Khaled

- Applied Principle 3 (Product) and Principle 8 (Self)
- Ensured product quality, reliability, testing, documentation, and personal professional development ethics
- Linked NFRs and FRs to the product-quality ethical principles

Judgment, Management – Yasseen Ahmed

- Applied Principle 4 (Judgment) and Principle 5 (Management)
- Added analysis on objectivity, integrity, management responsibility, and ethical decision-making
- Ensured proper judgment alignment with system monitoring, conflict resolution, and risk mitigation

Profession – Mohamed Mustafa

- Applied Principle 6 (Profession)
- Ensured professional responsibility, legal compliance, and ethical environmental behavior
- Documented compliance with standards, licensing, and academic integrity

Colleagues – Mohamed Adel

- Applied Principle 7 (Colleagues)
 - Covered collaboration ethics, fair treatment, knowledge sharing, and peer support
 - Integrated colleague-responsibility principles with team workflow
-

6.2 Ethical Challenges (Section 2)

1. Privacy – Mohamed Mustafa

Authored ethical analysis for data privacy, encryption, registration consent, and guest mode anonymity.

2. Inclusivity – Mohamed Mustafa

Evaluated balancing accessibility vs. system security, guest mode, multilingual support.

3. Environmental Validation – Mohamed Adel

Documented ethical importance of accurate material classification, contamination prevention, and AI validation integrity.

4. Gamification – Mohamed Adel

Evaluated fairness, behavioral motivation, and preventing unethical gaming of the system.

5. System Reliability – Ahmed Khaled

Analyzed reliability, uptime, monitoring, offline mode, failure handling, and safety concerns.

6. Transparency – Marawan Khaled

Handled transparency ethics: point calculation clarity, environmental impact reporting, accountability mechanisms.

7. Intellectual Property – Yasseen Ahmed

Documented licensing compliance, open-source usage ethics, and protection of third-party intellectual property.

6.3 Documentation & Legal Compliance (Section 3)

Lead: Mohamed Mustafa

- Verified licensing, attributions, and legal compliance
- Mapped ethics principles to documentation standards
- Ensured accurate referencing and copyright adherence

Support: Yasseen Ahmed

- Reviewed compliance with intellectual property principles
 - Ensured licensing alignment with ethical and legal standards
-

6.4 Ethical Challenges Resolution (Section 4)

Lead: Marawan Khaled

- Compiled system-wide resolutions
- Integrated transparency and product-quality considerations

Support: Ahmed Khaled & Mohamed Adel

- Supported reliability, validation, and fairness improvements
-

6.5 Report Integration & Quality Assurance

All Members

- Full-team peer review
- Cross-check of ethical principles
- Validation of requirement-to-ethics mapping
- Consistency checks for compliance with the ACM/IEEE Cod

6.7 Individual Contributions Summary

Team Member	Ethics Principles	Ethical Challenges	Additional Responsibilities
Ahmed Khaled	Public, Client & Employer	System Reliability	Alternatives analysis, requirement mapping
Mohamed Mustafa	Profession	Privacy, Inclusivity	Documentation & legal compliance
Marawan Khaled	Product, Self	Transparency	Challenge resolutions, product-quality mapping
Mohamed Adel	Colleagues	Environmental Validation, Gamification	Review and conclusion refinement
Yasseen Ahmed	Judgment, Management	Intellectual Property	Ethics framework, integration, quality assurance

All team members contributed equally to the project's ethical decision-making process, with each member bringing unique perspectives on privacy, accessibility, environmental impact, and system integrity. The collaborative approach ensured comprehensive coverage of ethical considerations and maintained consistency with professional software engineering standards.

Team Members:

- Yasseen Ahmed El-Sayed (23101608)
- Mohamed Mustafa Awad (23101594)
- Ahmed Khaled Said (23101545)
- Marawan Khaled Ahmed (23101599)
- Mohamed Adel Abdulrahman (23101899)

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